# Programming Fundamentals Using C++

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# Week 02

**Lecture – 01**

**Lecture – 02**

**Semester Calendar**

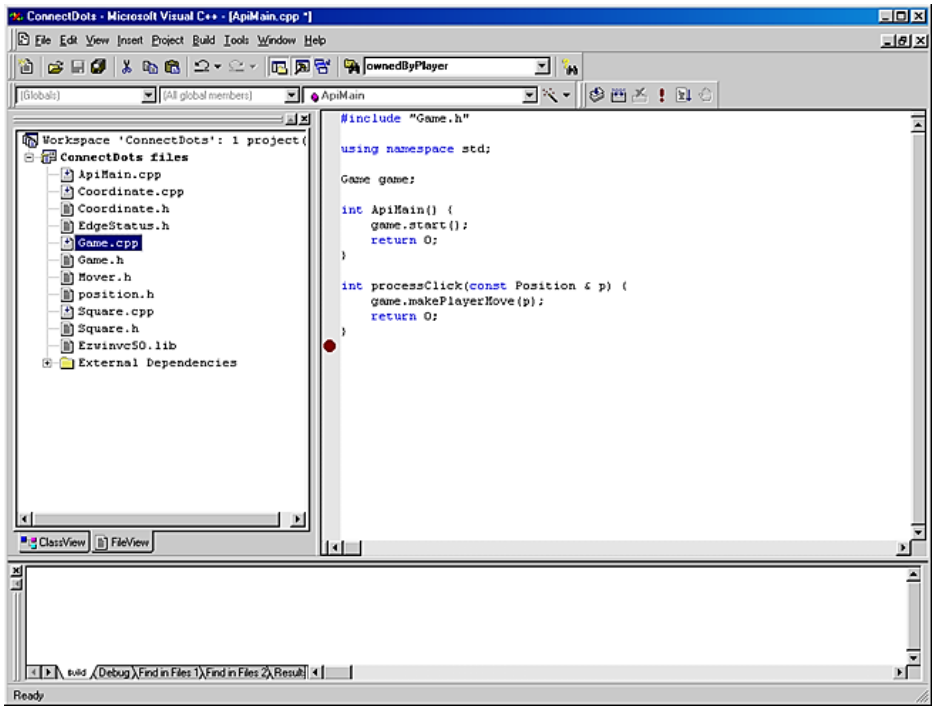
# Class Schedule

**Lab Hours (Practice)**

**Wednesday (09:30 to 12:30)**

# Basic Program Construction

- Compilers take source code and transform it into executable files, which your computer can run.
- Source files are text files (extension .CPP)
- Executable files have the .EXE extension



NetBeans

Codelite

IDE multiplataforma para C/C++

TURBO C++

eclipse

# IDE Advantages:

- Handles build process for you
- Syntax highlighting and live error detection
- Code completion (fills in as you type)
- Creation of files via templates
- Built-in debugging
- Code refactoring (ex. Change a variable name everywhere in your code)
- Higher productivity than plain text editors!

# Edit

- The first phase consist of editing a file.
- The programmer types a C++ program with editor.
- The program source file is then stored on secondary storage device such as disk.
- C++ program file names often end with .cpp

# Compile

- Next, the programmer gives the command to compile the program.
- Use the compiler to:
  - Check that the program obeys the language rules
  - Translate into machine language (object program)
- In C++ system
  - A preprocessor program executes automatically before the compiler's translation phase begins.
  - Preprocessor obeys commands called preprocessor directives.
    - Indicates that certain manipulations are to be performed on the program before compilation.

# Linking:

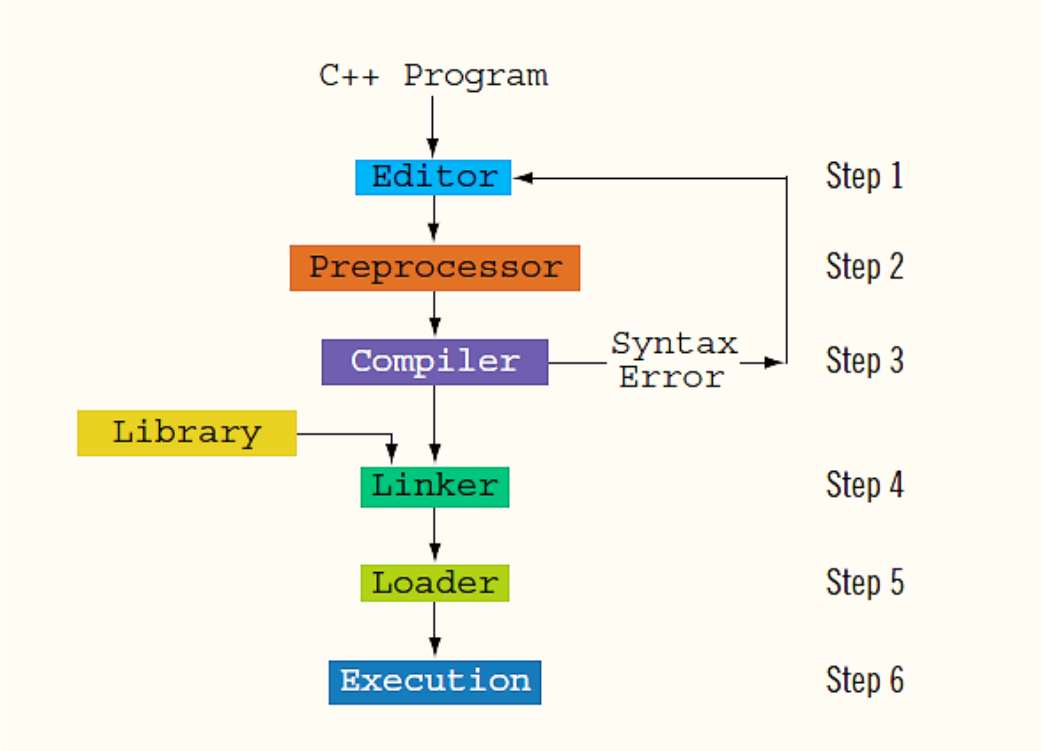
- The preprocessor is invoked by the compiler before the program is converted to machine language.
- The next phase is called linking.
  - A linker links the object code with the code for the missing functions to produce an executable image.
- If the program compiles and links correctly, an executable image is produced.
- Linker:
  - Combines object program with other programs provided by the SDK to create executable code
  - Library: contains prewritten code you can use

# Loader:

- The next phase is called loading.
- Before a program can be executed, the program must first be placed in memory. This is done by the loader, which takes the executable image from disk and transfers it to memory

# Execute:

- Finally, the computer executes the program one instruction at a time.
- Some IDEs do all this with a Build or Rebuild command



# A Very Brief History of C++

**1962**
Simula I was invented by Kristen Nygaard and Ole-Johan Dahl as a *simulation language*

**1967**
Simula 67 developed as the first *object-oriented* language

**1969-1973**
The C language was invented by Dennis Ritchie at Bell Labs

**1972**
D. Ritchie and Ken Thompson re-write the Unix OS in C

**1979**
Bjarne Stroustrop began developing "C with Classes"

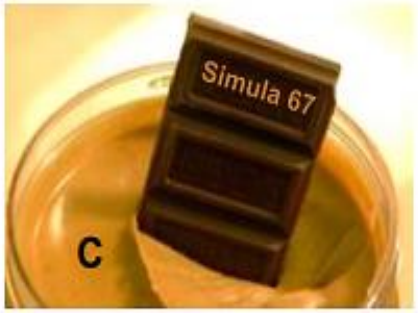
**1983**
"C with Classes" renamed to C++

**1985**
1st commercial C++ compiler, Cfront, released by AT&T

**1989**
C++ 2.0 standard released.

**2011**
Major update: C++11 standard released

**2014**
Minor update: C++14 released.



C++

For details more check out A History of C++: 1979−1991

# History of C and C++

- History of C
  - Evolved from two other programming languages
  - BCPL and B
    - “Typeless” languages
  - Dennis Ritchie (Bell Laboratories)
    - Added data typing, other features
  - Development language of UNIX
  - Hardware independent
    - Portable programs
  - 1989: ANSI standard
  - 1990: ANSI and ISO standard published
    - ANSI/ISO 9899: 1990

# History of Computer Languages:

- History of C++
  - Extension of C
  - Early 1980s: Bjarne Stroustrup (Bell Laboratories)
  - Provides capabilities for object-oriented programming
    - Objects: reusable software components
      - Model items in real world
    - Object-oriented programs
      - Easy to understand, correct and modify
  - Hybrid language
    - C-like style
    - Object-oriented style
    - Both

# Getting Started

**Building Program**

**&**

**Compiling**

# Structure of C++ Program

| Source Code | Output |
| --- | --- |
| // My First C++ Program<br>#include<iostream><br>using namespace std;<br>int main()<br>{<br>cout<<" ** Hello World **";<br><br>return 0;<br>} | ** Hello World ** |

# Components of C++ Program:

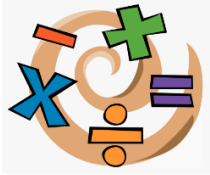
- Line 1: // My First C++ Program
- Lines Begin with two slash sign (//) are called comments by the programmer and have no effect on the behavior of the program .
- Programmer use them to include short observation or explanation concerning the code or program.
- In this case, It is a brief introductory description of the program.
- Note: One line Comment

# Components of C++ Program (Conti……)

- Line 2: *#include<iostream>*
- Lines Begin with hash sign (#) are called pre-processor directives or pre-compiler directives, include some standard library header files
- In this case, the directive *#include<iostream>,* instruct the pre-processor to include a section of standard c++ code , Known as header *iostream* , that allow standard input output operations.
- *iostream* makes input and output available to us
- *iostream* standard header file that defines various types and variables related to I/O

# Components of C++ Program (Conti….)

- Line 3: using namespace std;
- legalizes" header contents” .
- Use the standard namespace
- Loads a namespace called std. Namespaces are used to separate sections of code for programmer convenience. To save typing we’ll always use this line in this course.



# Components of C++ Program (conti….)

- Line 4: int main()
- int main() , begin function main()
- The line initiate the declaration of the main function.
- Essentially, a function is a group of instruction/statements which are given name.
- In this case, gives the name " main " to the group of statements/instructions that follows.
- The execution of all C++ programs begin with the main function regardless of where the function is actually located within the code.

**Note: open and close braces { } mark the beginning and end of main function**

# Components of C++ Program (Conti…)

- Line 5: cout<<“ ** Hello World ** ”;
- Prints the message ** Hello World ** .
- cout is an instance of iostream
- cout is the object that writes to the stdout device, i.e. the console window.
- It is part of the C++ standard library.
- Use with “using namespace std;”
- Operator << is used for output
- << is the C++ insertion operator. It is used to pass characters from the right to the object on the left.

# Components of C++ Program:

- Line 6: return 0;
- The return statement returns an integer value to the operating system after completion.
- 0 means “no error”. C++ programs must return an integer value.